

Problem Definition

- The CPU cooling system consists of an aluminum heat sink, a glass-filled nylon fan, and a plastic casing.
- The heat sink base experiences a steady heat flux of 2000 W/m^2 while the fan forces air at 28°C over the system with a convection coefficient of $30 \text{ W/m}^2\cdot^\circ\text{C}$.
- The fan rotates at 5000 RPM, creating additional structural loading due to centrifugal forces.
- Goal: Run steady-state thermal and thermo-structural analyses to determine temperature distribution, heat flow, stress levels, deformation, and safety factors for each component.
- A separate analysis of a $1/7$ sector of the fan will be used to study stress, deformation, and safety factor as rotational speed varies.
- A full-fan modal analysis will be performed to identify the first six natural frequencies and mode shapes and assess vibration risks.

Aluminum Alloy Material Properties

Properties of Outline Row 3: Aluminum Alloy			
	A	B	C
1	Property	Value	Unit
2	 Material Field Variables	 Table	
3	 Density	2700	kg m ⁻³ ▼
4	 Isotropic Secant Coefficient of Thermal Expansior		
5	 Coefficient of Thermal Expansion	2.36E-05	C ⁻¹ ▼
6	 Isotropic Elasticity		
7	Derive from	Young's Modulus an... ▼	
8	Young's Modulus	69800	MPa ▼
9	Poisson's Ratio	0.33	
10	Bulk Modulus	6.8431E+10	Pa
11	Shear Modulus	2.6241E+10	Pa
12	 Tensile Yield Strength	276	MPa ▼
13	 Isotropic Thermal Conductivity	167	W m ⁻¹ K ⁻¹ ▼

Data derived from
the following table:

Table-1: Material properties for aluminum	
Density (<i>kg/m³</i>)	2700
Module of Elasticity (Mpa)	69800
Poisson's ratio	0.33
Coefficient of thermal expansion (1/°C)	23.6E-6
Heat conduction coefficient (W/(m.K)	167
Yield strength (Mpa)	276

Glass-Filled Nylon Material Properties

Properties of Outline Row 4: Glass Filled Nylon			
	A	B	C
1	Property	Value	Unit
2	 Material Field Variables	 Table	
3	 Density	1360	kg m ⁻³
4	 Isotropic Secant Coefficient of Thermal Expansior		
5	 Coefficient of Thermal Expansion	2.56E-05	C ⁻¹
6	 Isotropic Elasticity		
7	Derive from	Young's Modulus an...	
8	Young's Modulus	6200	MPa
9	Poisson's Ratio	0.35	
10	Bulk Modulus	6.8889E+09	Pa
11	Shear Modulus	2.2963E+09	Pa
12	 Tensile Yield Strength	55	MPa
13	 Isotropic Thermal Conductivity	0.23	W m ⁻¹ K ⁻¹

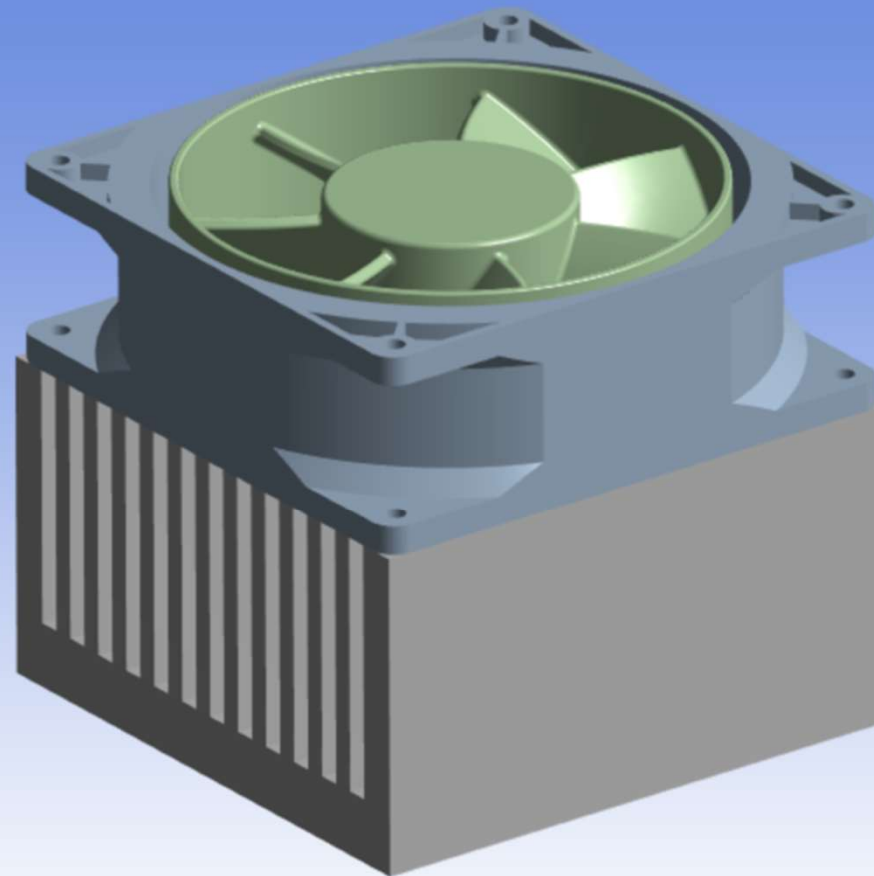
Data derived from
the following table:

Table-2: Material properties for glass-filled nylon	
Density (kg/m^3)	1360
Module of Elasticity (Mpa)	6200
Poisson's ratio	0.35
Coefficient of thermal expansion ($1/^\circ C$)	25.6E-6
Heat conduction coefficient (W/(m.K))	0.23
Yield strength (Mpa)	55

Model

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Model
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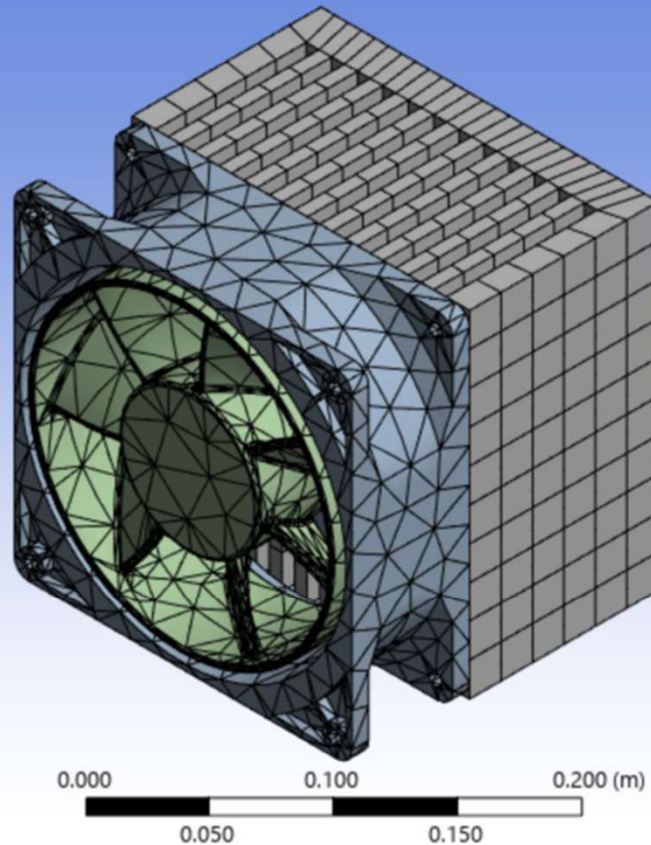


0.000 0.050 0.100 0.150 0.200 (m)



Model of fan and motor, fan casing, and heat sink

Mesh Quality and Quantity



Details of "Body Sizing" - Sizing	
Scope	
Scoping Method	Geometry Selection
Geometry	3 Bodies
Definition	
Suppressed	No
Type	Element Size
☐ Element Size	2.e-002 m
Advanced	
☐ Defeature Size	Default
Behavior	Soft

Statistics	
☐ Nodes	28869
☐ Elements	12594
Show Detailed Statistics	No

Meshing of all three bodies.

It should be noted that the number of nodes and elements were reduced due to simulations being run on a student license.

Steady State Thermal - Boundary Conditions

A: Steady-State Thermal

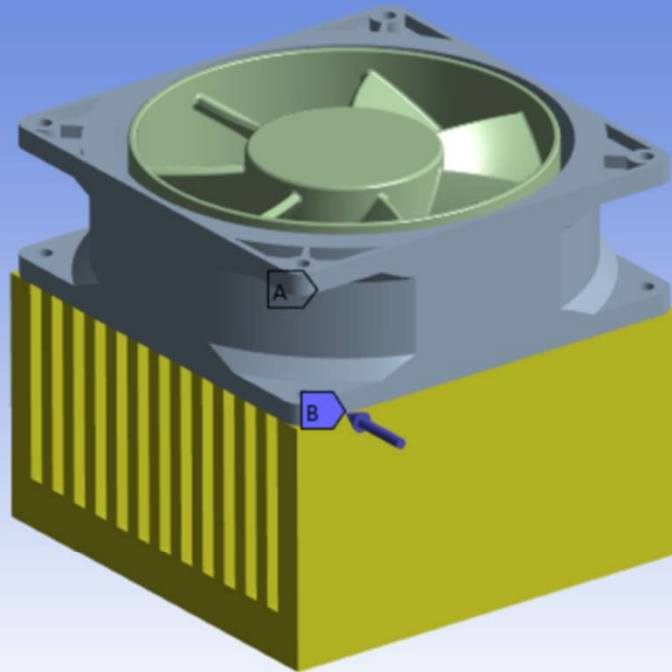
Heat Flux

Time: 1. s

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A Convection: 28. °C, 30. W/m²·°C

B Heat Flux: 2000. W/m²

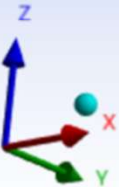
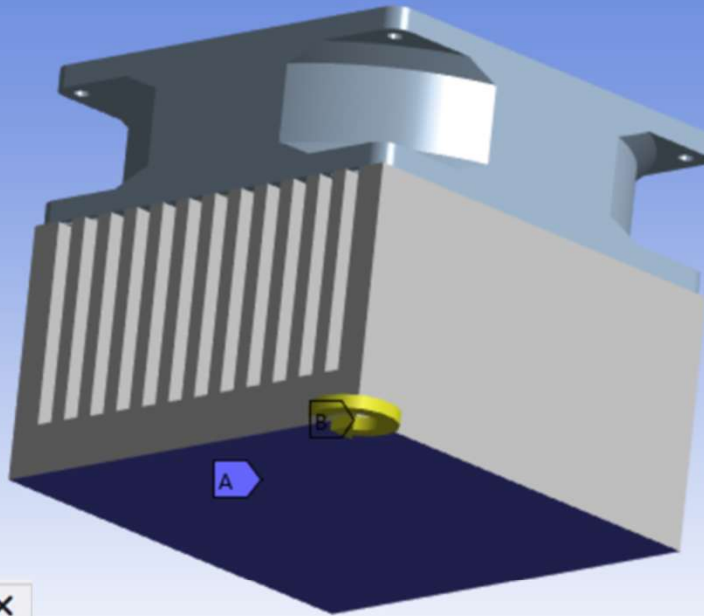


Details of "Initial Temperature"	
Definition	
Initial Temperature	Uniform Temperature
Initial Temperature Value	22. °C

Static Structural - Boundary Conditions

B: Static Structural
Fixed Support
Time: 1. s
12/12/2025 11:38:42 AM

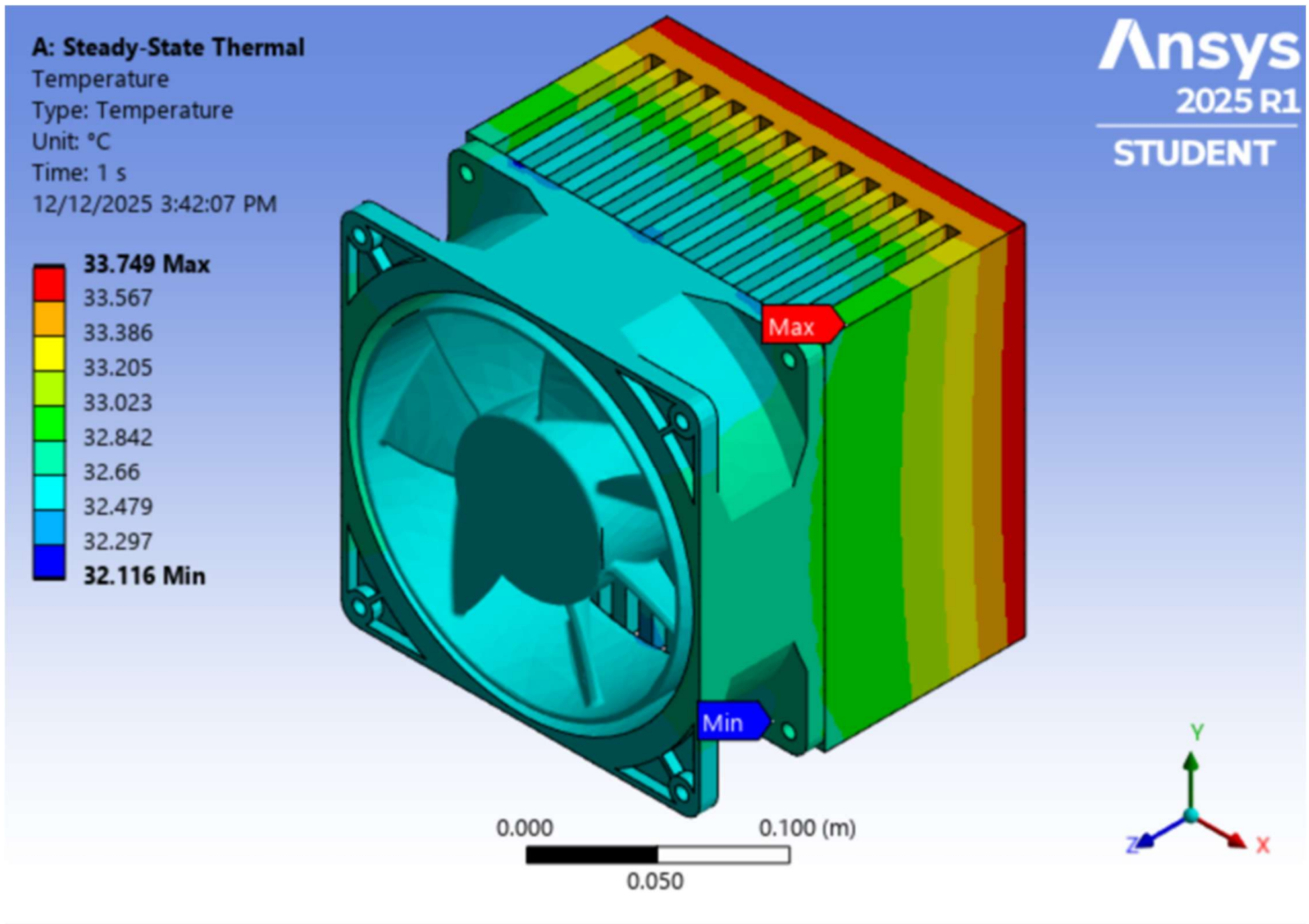
A Fixed Support
B Rotational Velocity



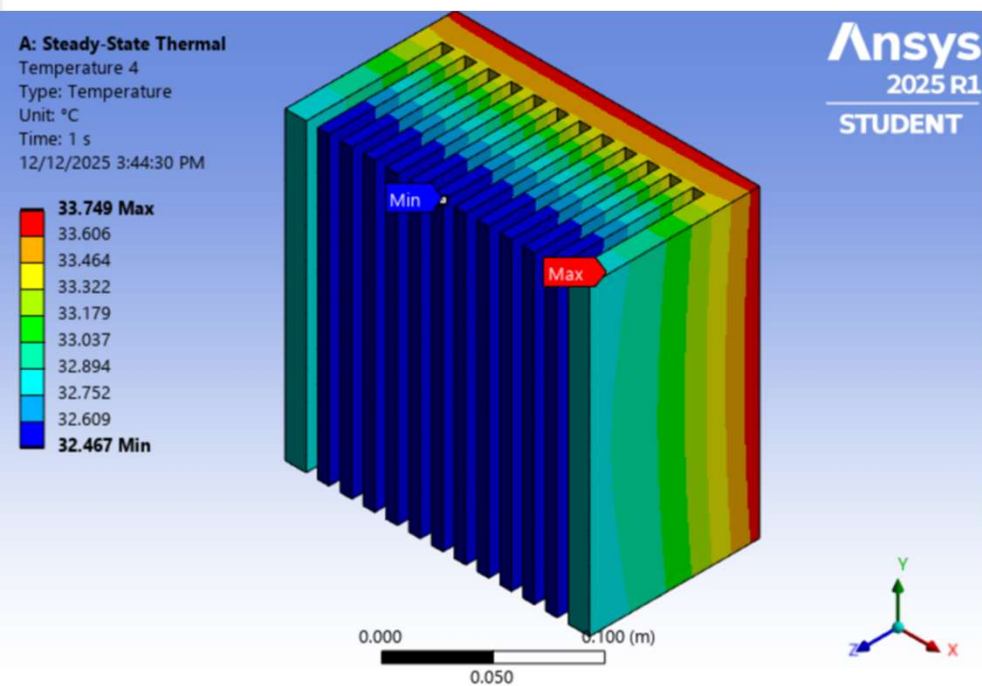
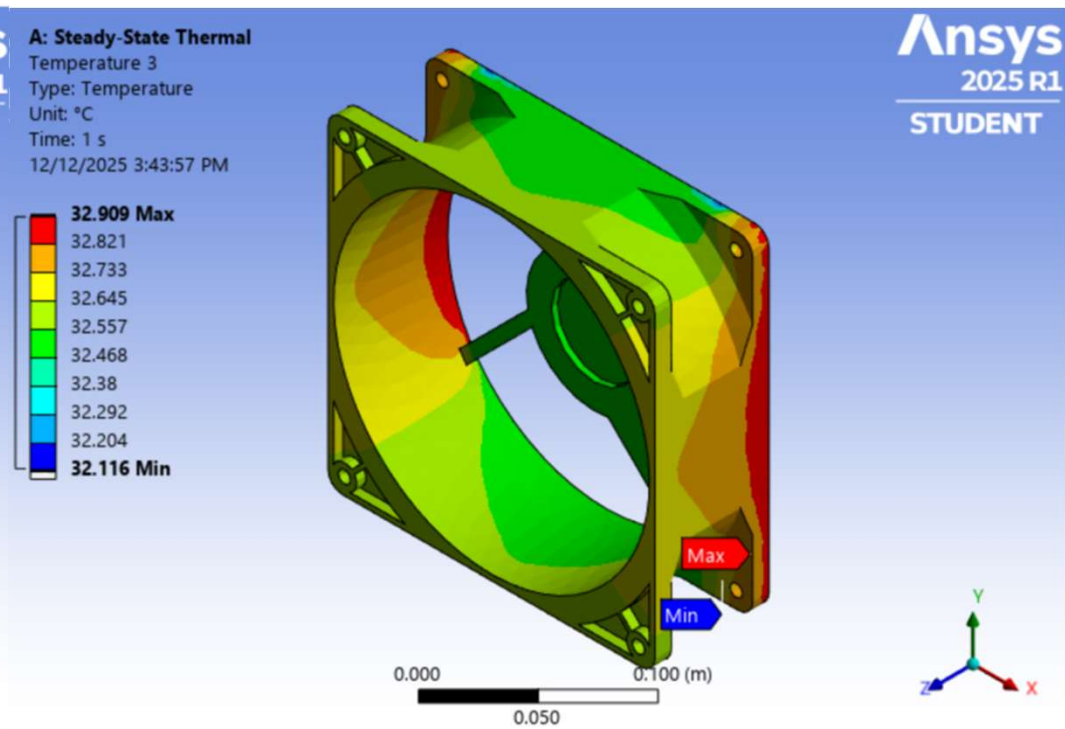
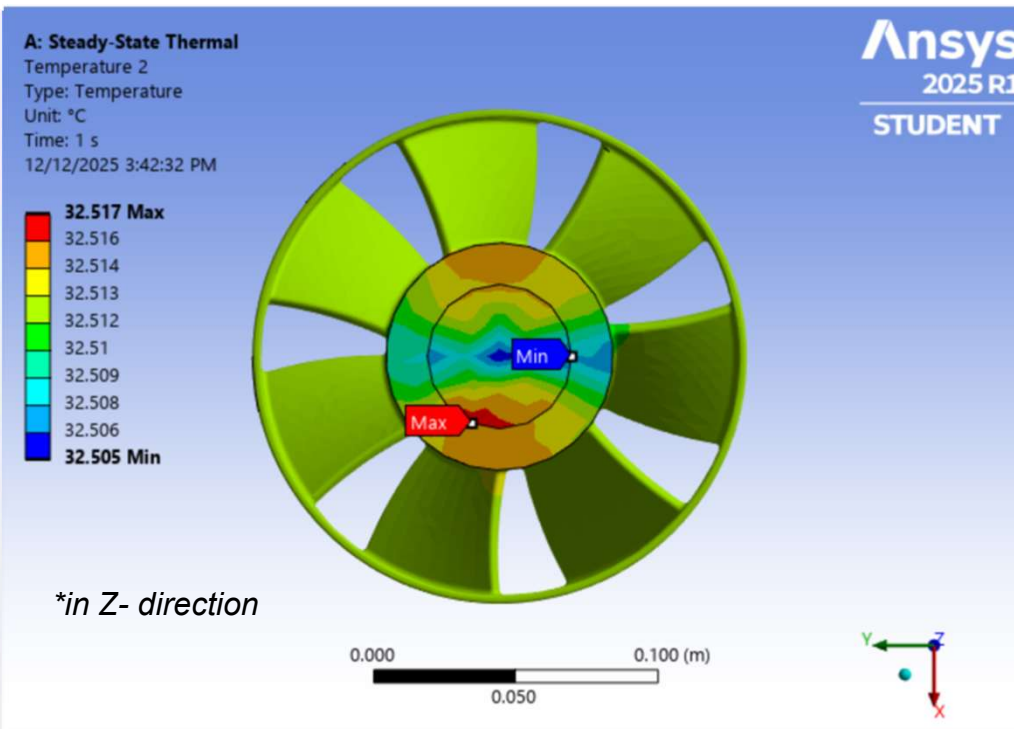
Details of "Rotational Velocity"	
Scope	
Scoping Method	Geometry Selection
Geometry	1 Body
Definition	
Define By	Components
Coordinate System	Global Coordinate System
☐ X Component	0. rad/s (ramped)
☐ Y Component	0. rad/s (ramped)
☒ Z Component	523.6 rad/s (ramped)
☐ X Coordinate	0. m
☐ Y Coordinate	0. m
☐ Z Coordinate	0. m
Suppressed	No



Steady State Thermal Contour Plot for All Bodies

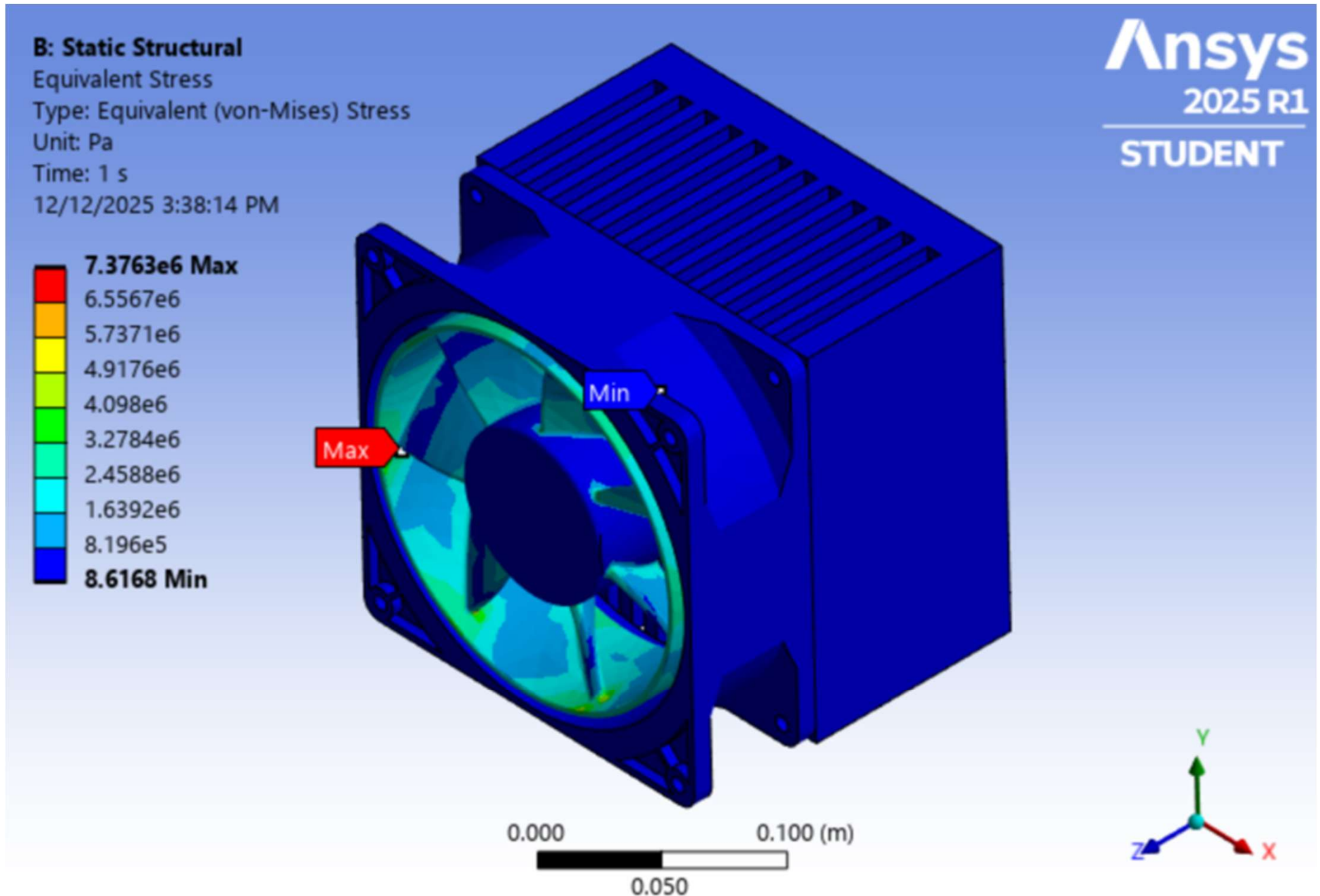


Steady State Thermal Contour Plots and Answers

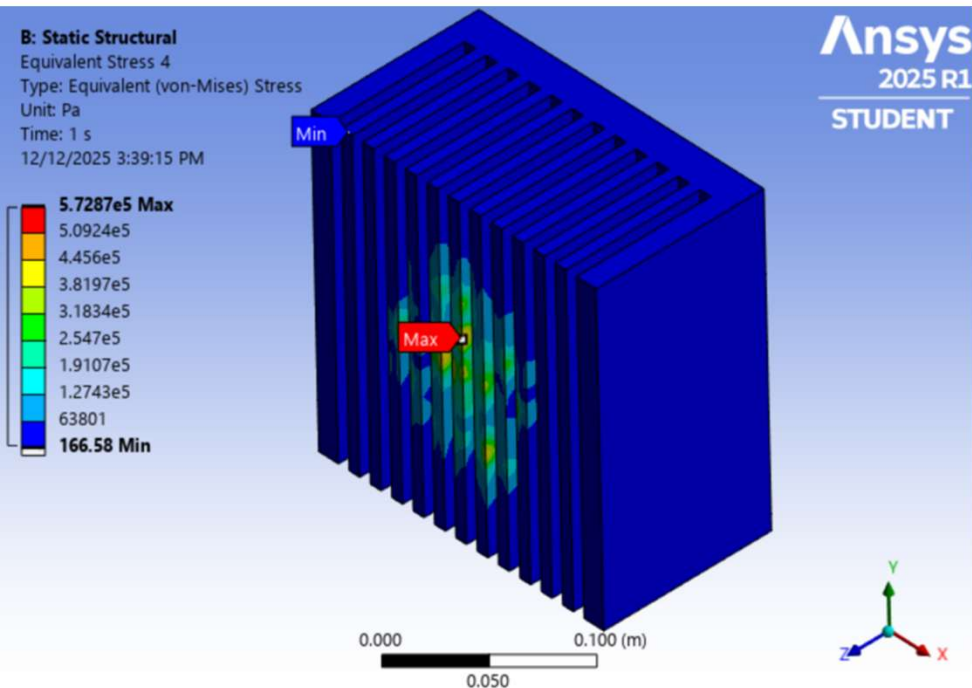
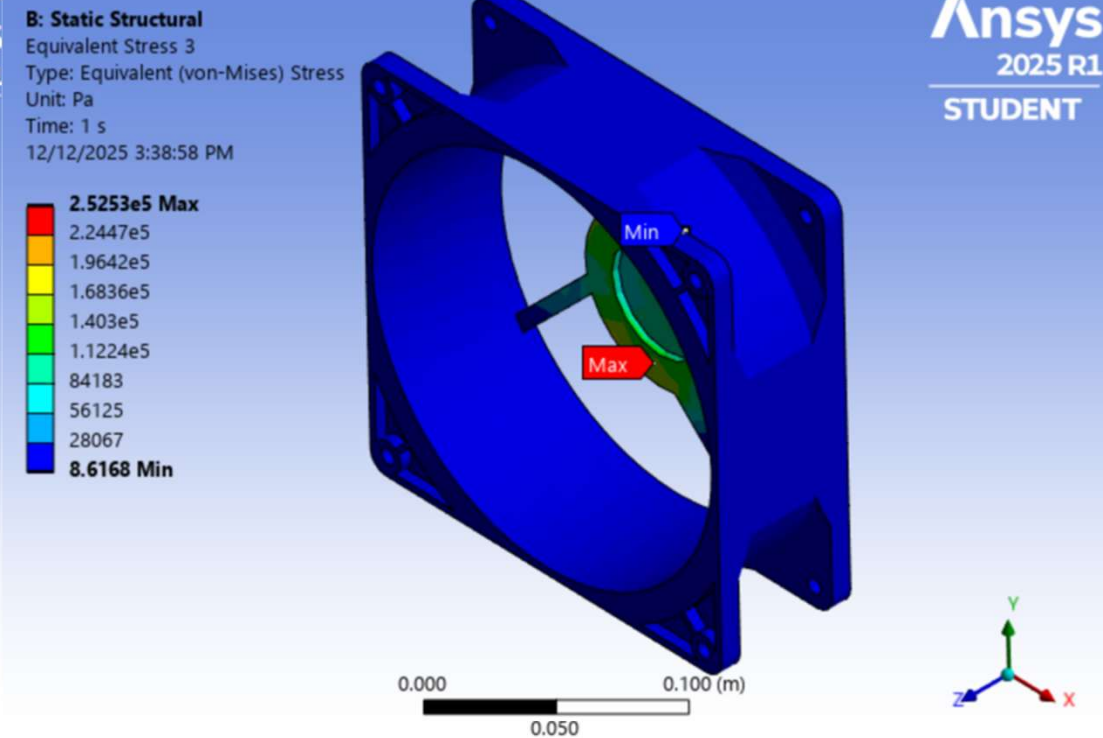
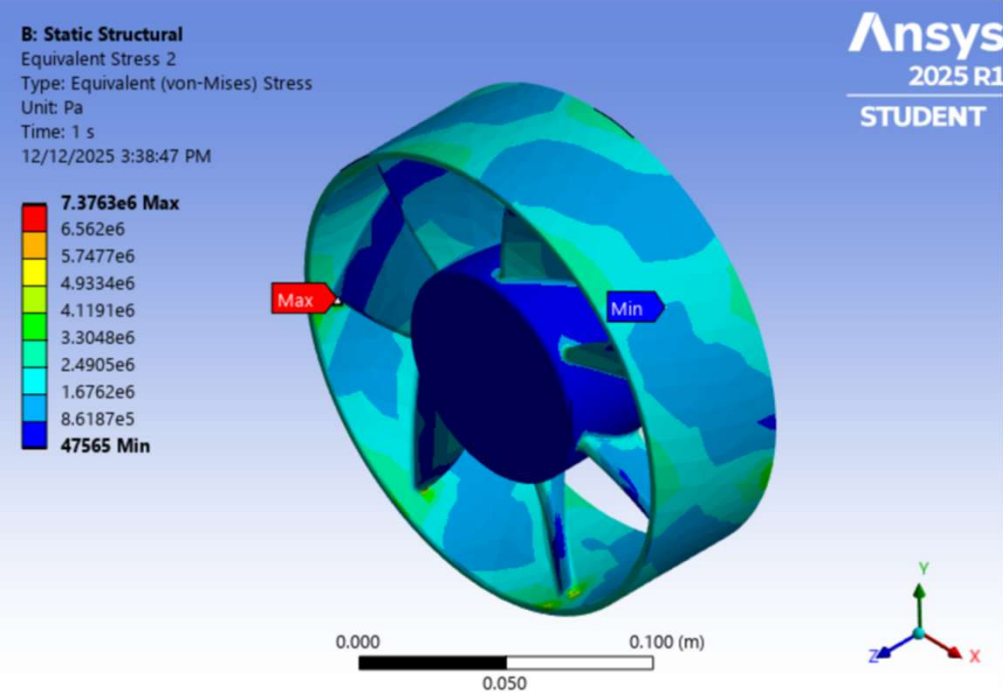


Heat drops off quickly at the aluminum–plastic interface because aluminum conducts heat very well while glass-filled nylon conducts it very poorly. The large difference in thermal conductivity creates a bottleneck, so heat spreads efficiently through the heat sink but slows down sharply once it reaches the casing and fan.

Static Structural Contour Plot for All Bodies

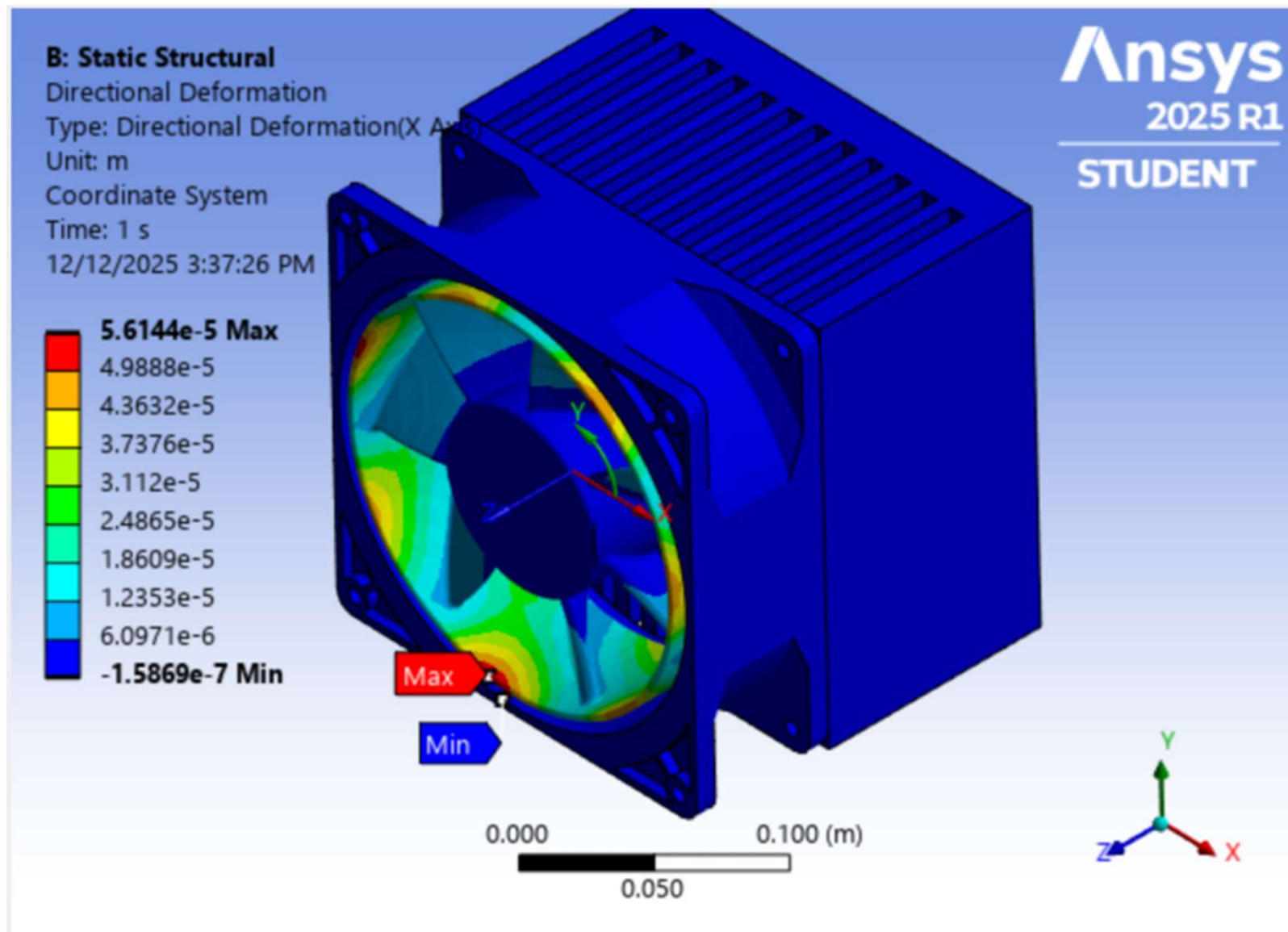


Static Structural Contour Plots and Answers



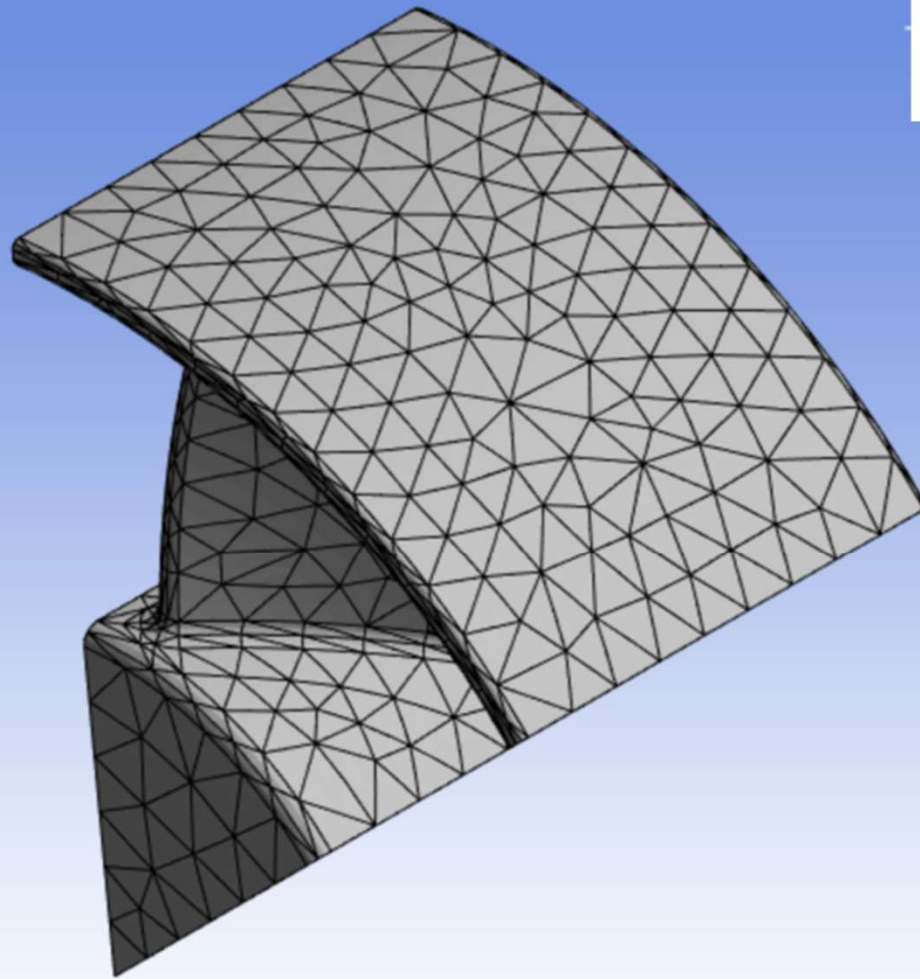
The fan shows the highest stress, reaching about 7.37×10^6 Pa, which is well above the peak stresses in the casing and heat sink. Even so, this value is far below the glass-filled nylon yield strength ($\approx 5.5 \times 10^7$ Pa), and the heat sink and casing are also operating well within their material limits. Based on these results, none of the components appear to be at risk of failure during normal operation.

Radial Deformation Contour Plots and Answers

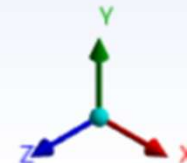


The deformation is far too small to close the tip gap. The plot shows a maximum directional deformation of about $5.6 \times 10^{-5} \text{ m}$ ($\approx 56 \text{ }\mu\text{m}$, 0.056 mm). Unless the design gap between the blade tip and casing is $\leq \sim 0.06 \text{ mm}$, this radial growth will not make the tip hit the casing

Mesh of 1/7 th sector of the fan



0.000 0.020 0.040 (m)



Statistics

☐ Nodes	6545
☐ Elements	3408
Show Detailed Statistics	No

Details of "Body Sizing" - Sizing

Scope

Scoping Method	Geometry Selection
Geometry	1 Body

Definition

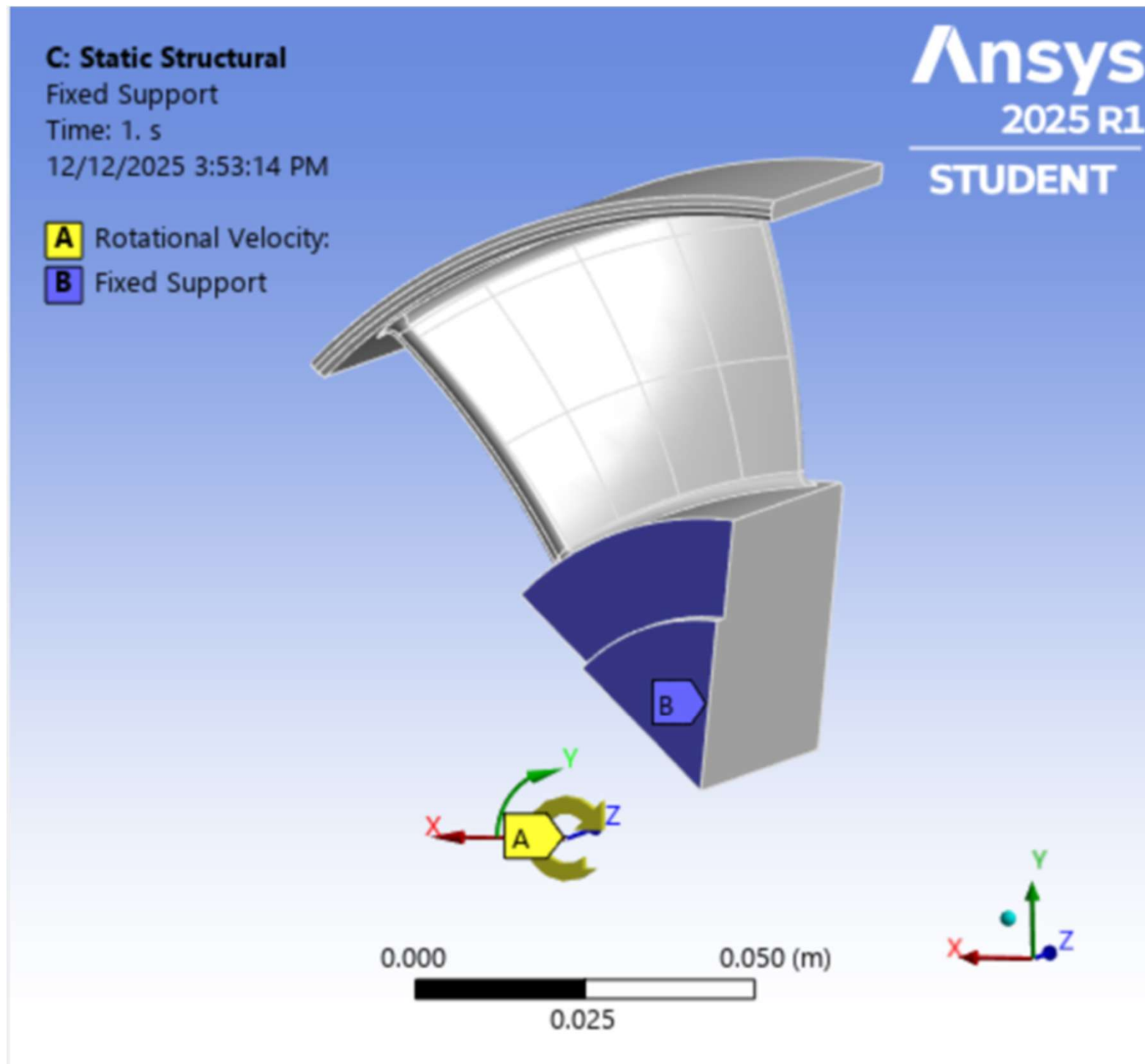
Suppressed	No
Type	Element Size
☐ Element Size	Default (6.4948e-003 m)

Advanced

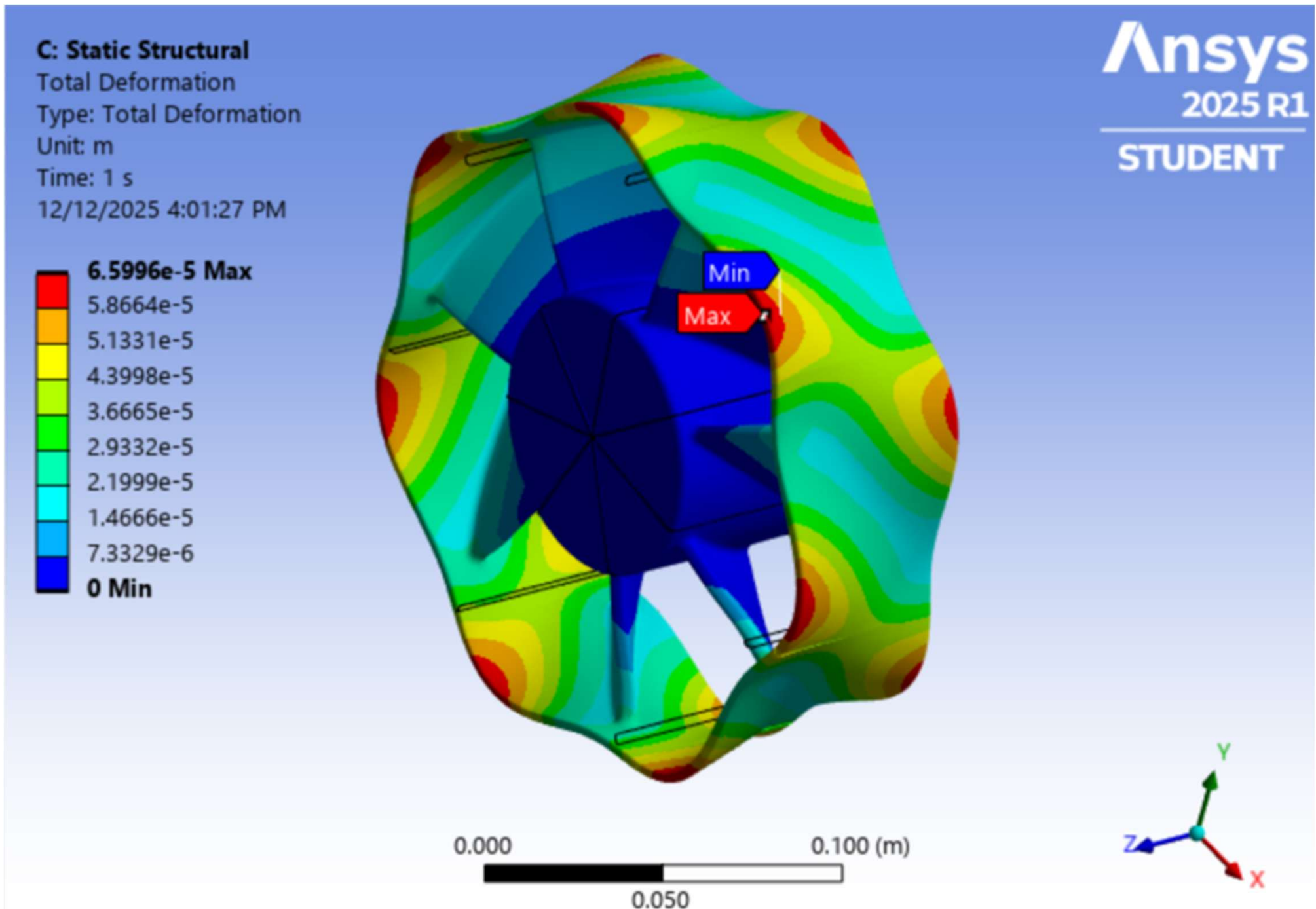
☐ Defeature Size	Default
Behavior	Soft

Meshing of 1/7 th sector of the fan. It should be noted that the number of nodes and elements were reduced due to simulations being run on a student license.

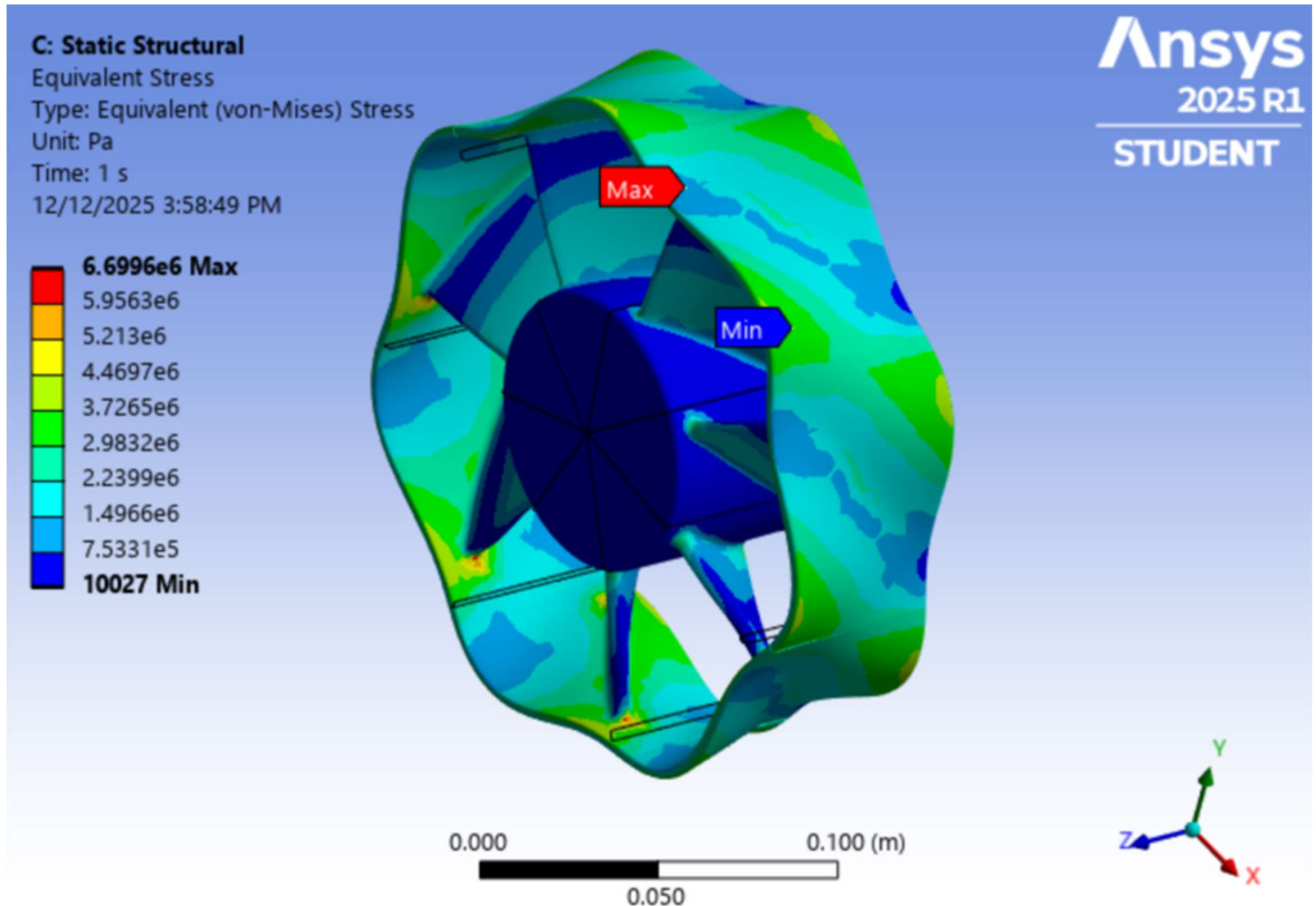
Steady State Thermal - Boundary Conditions



Static Structural Contour of Total Deformation



Static Structural Contour of Von-Mises Stress



Static Structural Contour of Radial Deformation

C: Static Structural

Directional Deformation

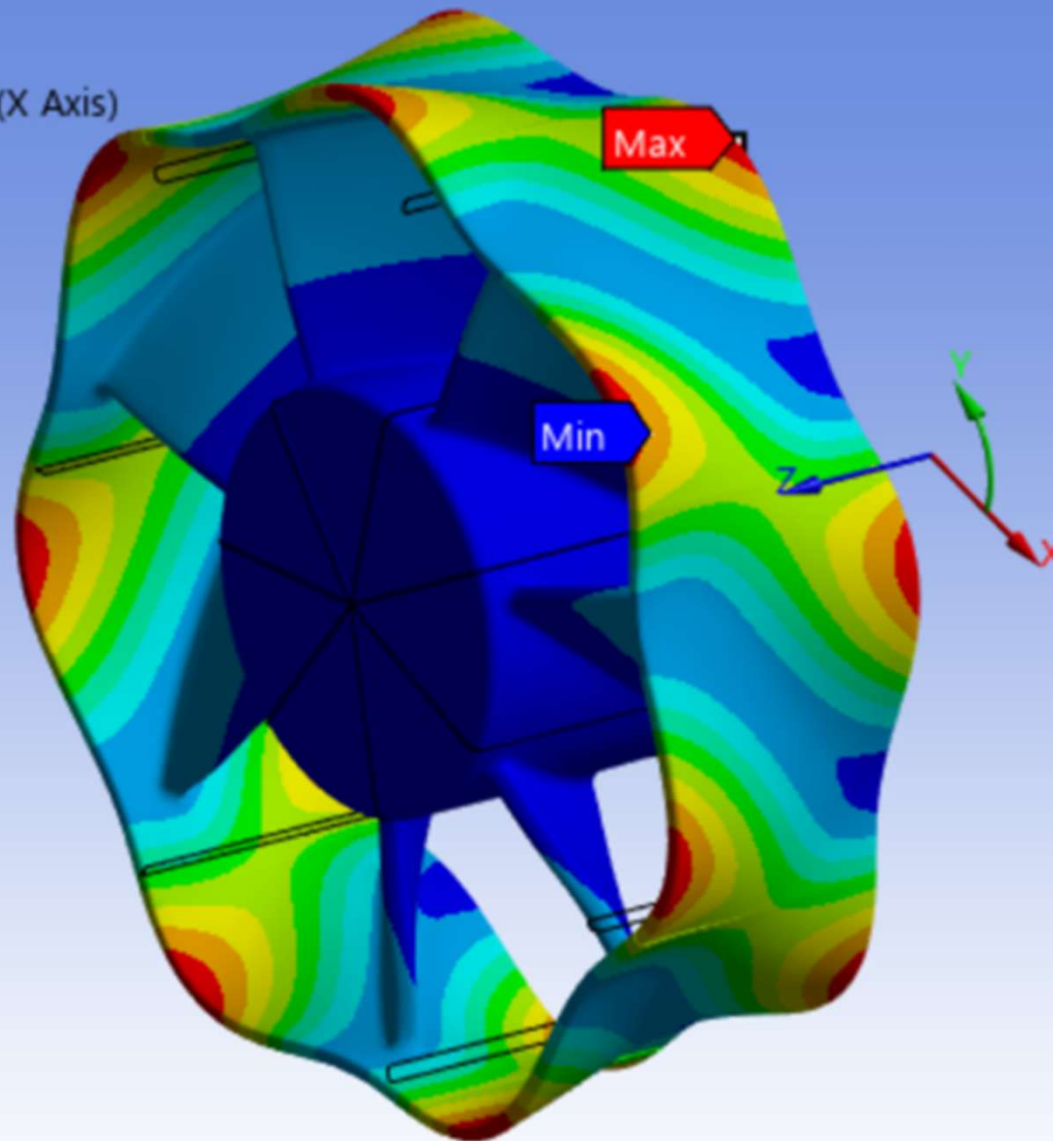
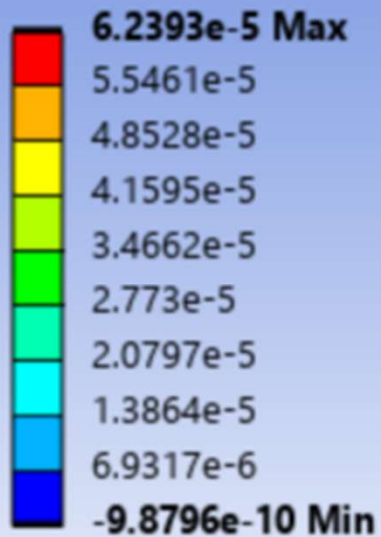
Type: Directional Deformation(X Axis)

Unit: m

Coordinate System

Time: 1 s

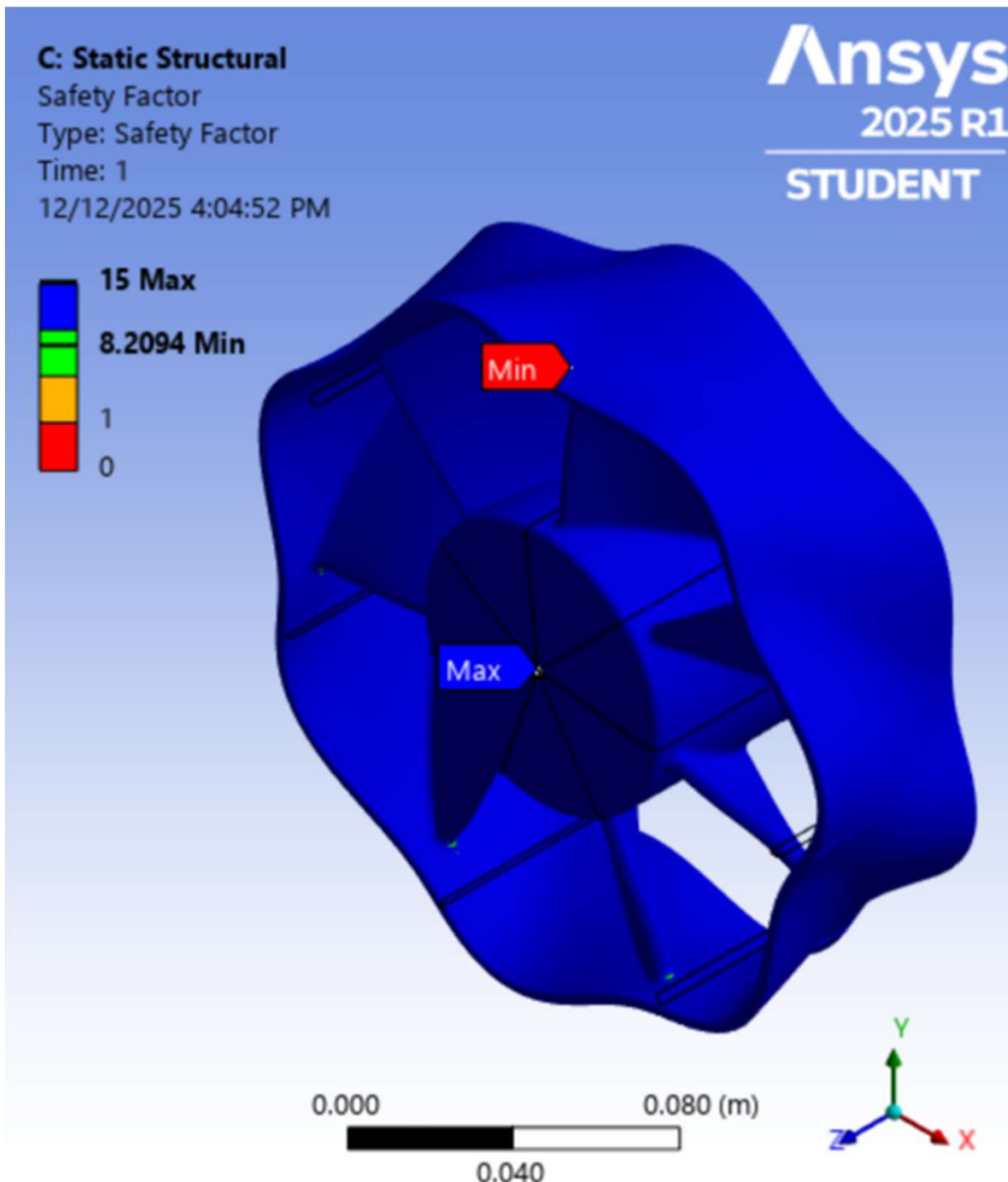
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Static Structural Contour of Safety Factor



The 1/7 sector analysis shows nearly the same behavior as the full-fan model: the fan reaches similar stress levels, deformation, and safety factor when the rotational load is applied. Without the temperature load from Part 1, the fan experiences slightly lower deformation and stress, but the difference is less than $1e6$ because the centrifugal forces are the primary response. Thermal expansion contributes only a little amount of additional radial growth compared to rotation, so removing the temperature load does change the overall stress state or the safety factor, just not by a lot.

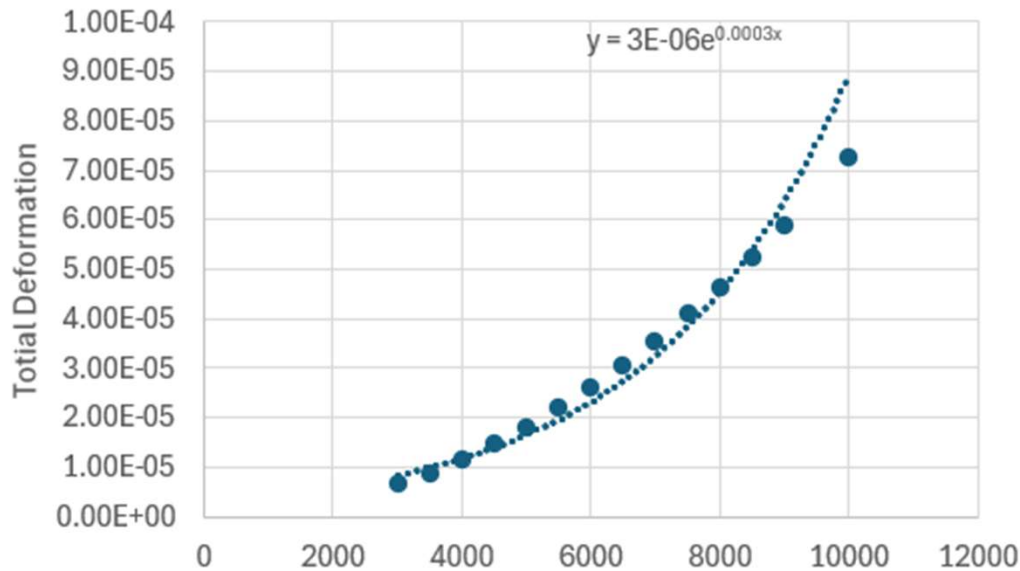
Parametric Model Data

Name	P1 - Rotational Velocity Z Component	P2 - Total Deformation Average	P3 - Safety Factor Minimum	P4 - Equivalent Stress Average
Units	rev min ⁻¹	m		Pa
DP 0 (Current)	5000	1.8192E-05	8.2094	1.429E+06
DP 1	3000	6.549E-06	15	5.1443E+05
DP 2	4000	1.1643E-05	12.827	9.1454E+05
DP 3	6000	2.6196E-05	5.701	2.0577E+06
DP 4	7000	3.5656E-05	4.1885	2.8008E+06
DP 5	8000	4.6571E-05	3.2068	3.6582E+06
DP 6	4500	1.4735E-05	10.135	1.1575E+06
DP 7	5500	2.2012E-05	6.7847	1.7291E+06
DP 8	3500	8.9139E-06	15	7.002E+05
DP 9	6500	3.0744E-05	4.8577	2.415E+06
DP 10	7500	4.0931E-05	3.6487	3.2152E+06
DP 11	8500	5.2574E-05	2.8407	4.1297E+06
DP 12	9000	5.8941E-05	2.5338	4.6299E+06
DP 13	10000	7.2767E-05	2.0524	5.7159E+06

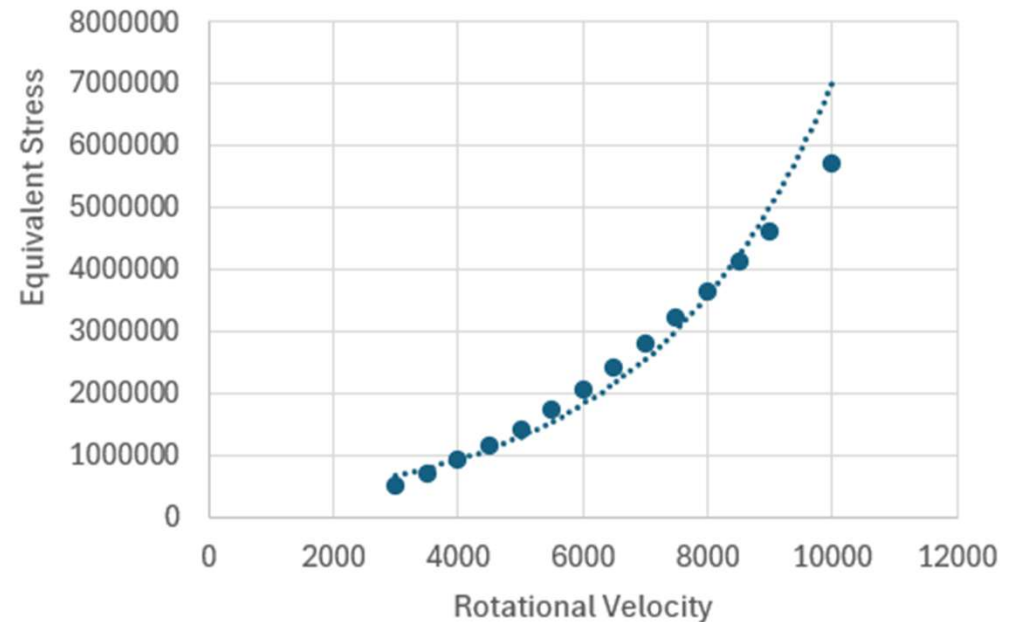
ID	Parameter Name	Value	Unit
Input Parameters			
Static Structural (C1)			
P1	Rotational Velocity Z Component	5000	rev min ⁻¹
New input parameter	New name	New expression	
Output Parameters			
Static Structural (C1)			
P2	Total Deformation Average	1.8192E-05	m
P3	Safety Factor Minimum	8.2094	
P4	Equivalent Stress Average	1.429E+06	Pa
New output parameter		New expression	
Charts			

Parametric Model Graphs

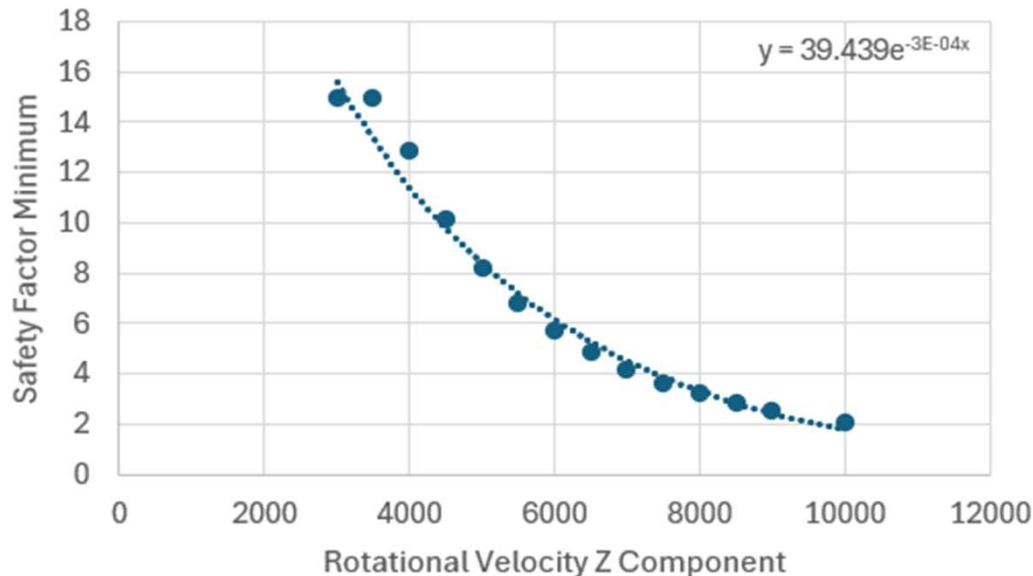
Plotted Rotational Velocity vs Total Deformation Average



Plotted Rotational Velocity vs Equivalent (Von-Mises) Stress Average



Plotted Rotational Velocity vs Safety Factor



These plots show a clear nonlinear, upward-curving trend for both total deformation and equivalent stress as rotational speed increases. This matches what you expect from linear elasticity, because the load created by rotation is proportional to the square of the angular velocity.

Parametric Model Solutions

If safety factor equaled 1.5, we would have a maximum rotational velocity the fan can be run at of approximately 10898 RPM.



$$y = 1.5$$

$$y = 39.439e^{(-3 \cdot 10^{-4})x}$$

$$\ln \frac{y}{39.439} = x$$

$$\frac{\left(\ln \frac{y}{39.439} \right)}{-3 \cdot 10^{-4}} = x$$

Name ▾	P1 - Rotational Velocity Z Component ▾	P2 - Total Deformation Average ▾	P3 - Safety Factor Minimum ▾	P4 - Equivalent Stress Average ▾
DP 14	10898	8.6417E-05	1.7282	6.7881E+06

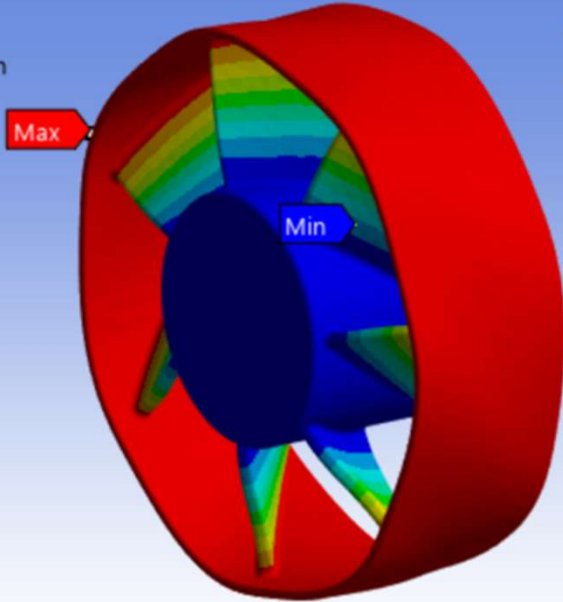
At a minimum safety factor of 1.5, the fan can run up to 10898 RPM. At this speed it reaches a total deformation of about 8.64×10^{-5} m, with most of that coming from radial growth. This deformation is far smaller than a typical fan-tip clearance, so there's no risk of the fan contacting the casing at this rotational velocity.

Modal Analysis of the Fan Component Plots 1-4

D: Modal

Total Deformation
Type: Total Deformation
Frequency: 443.03 Hz
Unit: m
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2.5731 Max
2.2872
2.0013
1.7154
1.4295
1.1436
0.85772
0.57181
0.28591
0 Min



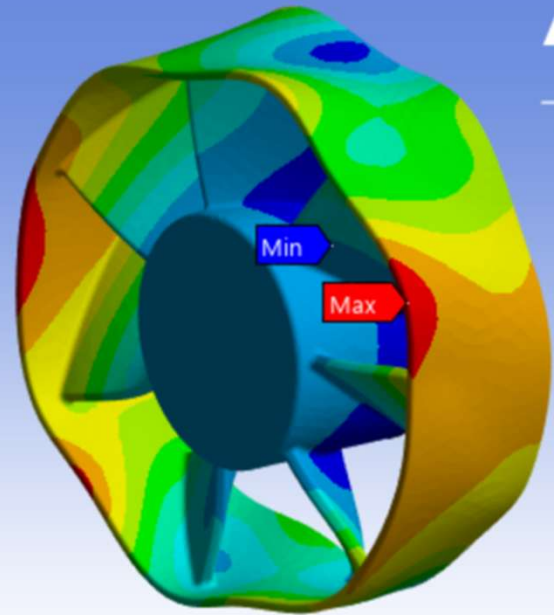
0.000 0.100 (m)
0.050

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D: Modal

Total Deformation 2
Type: Total Deformation
Frequency: 996.09 Hz
Unit: m
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3.8991 Max
3.4659
3.0327
2.5994
2.1662
1.7329
1.2997
0.86647
0.43324
0 Min



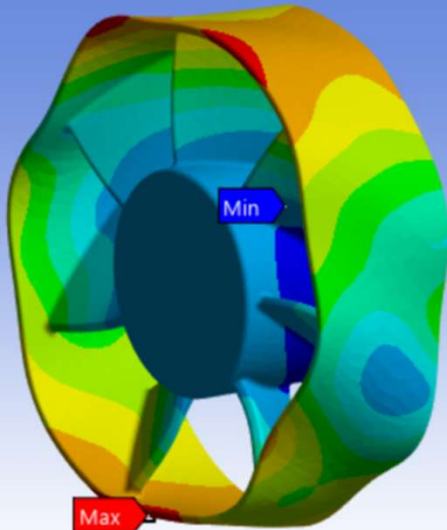
0.000 0.100 (m)
0.050

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D: Modal

Total Deformation 3
Type: Total Deformation
Frequency: 996.72 Hz
Unit: m
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3.9006 Max
3.4672
3.0338
2.6004
2.167
1.7336
1.3002
0.86679
0.4334
0 Min



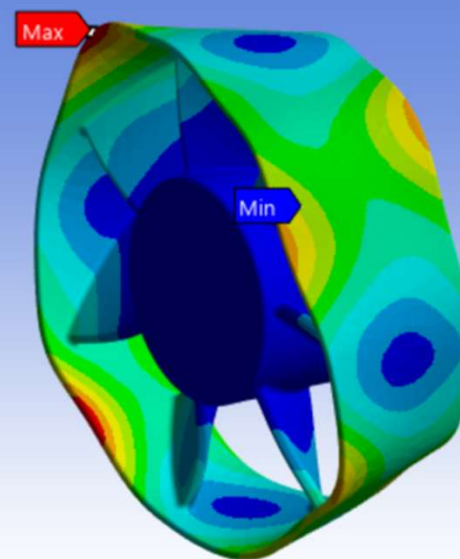
0.000 0.100 (m)
0.050

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D: Modal

Total Deformation 4
Type: Total Deformation
Frequency: 1280.8 Hz
Unit: m
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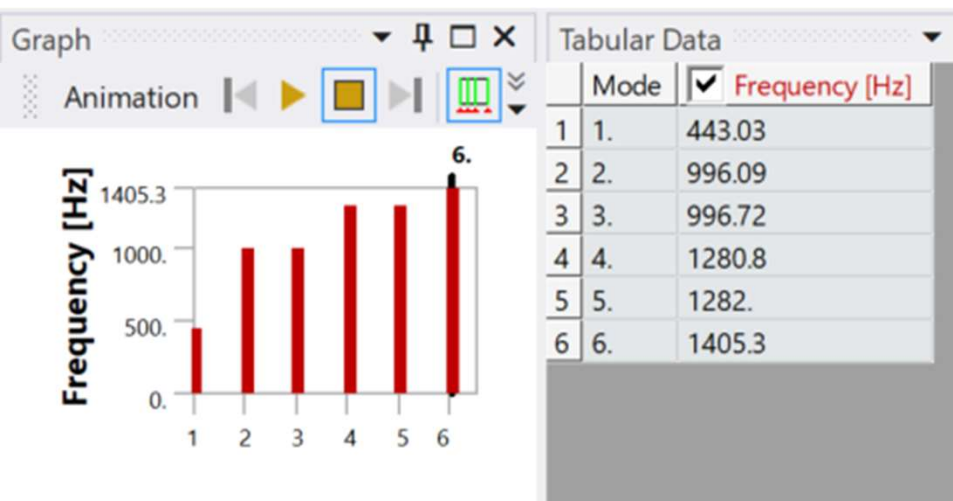
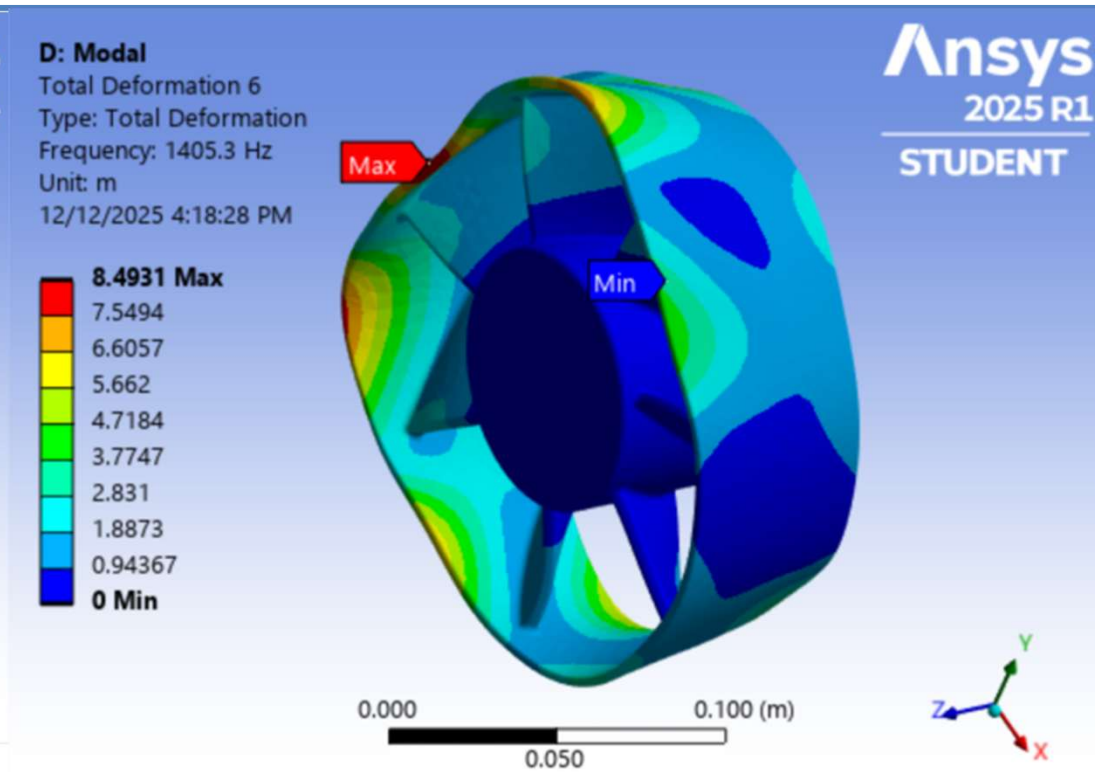
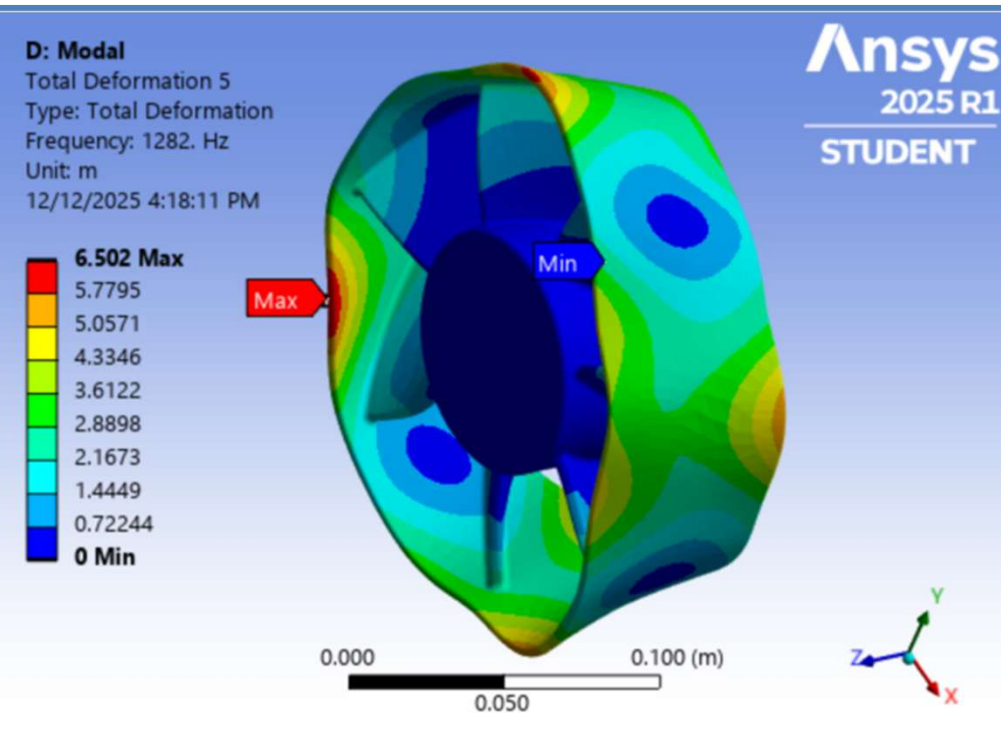
6.4164 Max
5.7034
4.9905
4.2776
3.5647
2.8517
2.1388
1.4259
0.71293
0 Min



0.000 0.100 (m)
0.050

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Modal Analysis of the Fan Component Plots 4-6



To cut the natural frequency of the fan component in half while keeping the component's geometry and material (design and stiffness) the same, the simplest physical change is to quadruple its mass. This is because the natural frequency is inversely proportional to the square root of the mass.

Conclusion

Across the full CPU cooling system and the fan-only models, the results show that the components operate well within their material limits under the applied thermal and rotational loads. The heat sink manages the imposed heat flux effectively, and the plastic casing and fan see much lower heat transfer due to their lower conductivity. Structurally, the fan carries the highest stress, but even at high rotational speeds its safety factor remains comfortably above one, and the radial deformation is far too small to close the tip gap or contact the casing. The 1/7 sector analysis confirmed the same trends while offering slightly cleaner results due to improved meshing. The parametric study showed the expected nonlinear increase in stress and deformation with RPM, consistent with centrifugal loads that scale with ω^2 . Finally, the modal analysis provides the natural frequencies needed to assess vibration risks. Overall, the system shows no indication of mechanical or thermal failure under normal operating conditions.